

In this paper, we investigate structure of group of small order.

Let G be an any group, and denote $|G|$ as order of G .

There are many tools to investigate the finite groups

⌈ Not necessary to consider the order.

- Lagrange's Theorem.
- Normalizer, Centralizer.

⌈ Necessary order consideration

- Class Equation
- Sylow Theorem.

$|G|=1$. It is just trivial.

$|G|=2$. Let $G=\{1, a\}$. If $a^2 \neq 1$, then $a^2=a$, therefore $1 \cdot a=a$ and $a \cdot a=a$, it means a has no inverse element. So, $a^2=1$, it means $G \cong \mathbb{Z}_2$.

$|G|=3$. Let $G=\{1, a, b\}$.